

2.1.2 FLAC

flac.sourceforge.net

FLAC, or Free Lossless Audio Codec, is another open source audio codec, but unlike MP3, Vorbis or others, it's lossless compression. This means you can rip CD audio without losing any data or quality whatsoever.

Let's quickly explain the term lossless compression: the simplest example is when you use WinZip or WinRAR to compress a file or multiple files: you can get back the original files when you decompress the archive, without losing any data whatsoever. Similarly, when you use lossless codecs such as FLAC, none of the digital data is lost—it's like zipping a CD audio file and then unzipping it during playback.

FLAC is an open source project that can be found at SourceForge.net. It can be used to compress (in a lossless way) audio, and used with the Ogg container to give you Ogg FLAC—or FLAC-encoded Ogg files.

In order to illustrate this, think of an audio file as consisting of two layers: the inner layer is the uncompressed digital audio data (FLAC-encoded audio), while the outer layer is the holder or “container” of this data. The container is what holds the audio data, broken and arranged into smaller parts, which allows you to seek (move the slider) through the song, or even for editing.

The advantage of using FLAC to encode your collection is that it is quite fast, both when encoding and decoding, and is natively seekable and streamable, so you don't really need a container like Ogg (unless you want to use your files with a portable music player that supports Ogg).

Overall, FLAC is good for audiophiles who want CD-quality data wherever they go. For most of us though, lossy formats such as MP3 and Ogg should do just fine. After all, most of us want compressed music to fit onto our portable media devices or cell